BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: MRD-10-240522 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. If $x^2 - x - 1 = 0$, then x = ...

(A) $(-1 \pm \sqrt{5})/2$
(B) $-1 \pm \sqrt{5}/2$
(C) $(1 \pm \sqrt{5})/2$
(D) $1 \pm \sqrt{5}/2$

Q2. Product of the roots of $3x^2 + 2x - 5 = 0$ is .....

(A) $-5/3$
(B) $2/3$
(C) $-2/3$
(D) $+5/3$

Q3. If $m \propto 1/n$ then .....

(A) m = kn
(B) n = km
(C) m/n = k
(D) mn = k

Q4. If 7 : 9 :: x : 27 .....

(A) x = 21
(B) x = 3
(C) x = 7
(D) x = 81

Q5. If P(x) and Q(x) are two polynomials then P(x)/Q(x), $Q(x) \neq 0$ is .....

(A) Irrational fraction
(B) Rational fraction
(C) Proper fraction
(D) Improper fraction

Q6. If 'f' is a function from A to B, then 'f' is onto function, if .....

(A) Range f = B
(B) Range f ≠ A
(C) Dom f = A
(D) None of these

Q7. If R = {(0,0), (8,2), (10,3), (14,12)} then DomR = ...

(A) {0,8,10,14}
(B) {0,2,3,12}
(C) {8,10,4}
(D) {0,10}

Q8. The number of times a value appears on a set of data is called .....

(A) Average
(B) Frequency
(C) Mode
(D) Median

Q9. In a set of data 41, 43, 47, 51, 57, 52, 59 median is .....

(A) 51
(B) 47
(C) 57
(D) None of these

Q10. cot θ = ...

(A) sinθ/cosθ
(B) 1/cosθ
(C) cosθ/sinθ
(D) 1/sinθ

Q11. 45° = ... radian.

(A) π/3
(B) π/4
(C) π/6
(D) π/2

Q12. Pythagoras was a ..... mathematician.

(A) Indian
(B) Pakistani
(C) Chinese
(D) Greek

Q13. Tangents drawn at the end points of the diameter of circle are .....

(A) Parallel
(B) Perpendicular
(C) Intersecting
(D) None of these

Q14. The angle in a semi-circle is a ..... angle.

(A) Acute
(B) Right
(C) Obtuse
(D) None of these

Q15. A cumulative frequency histogram is also called .....

(A) Histogram
(B) Pie chart
(C) Ogive
(D) Frequency polygon

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 36

Q2. Attempt any nine (9) of the following parts. All parts carry equal marks.

(i) Solve the equation: $6x^4 - 13x^2 + 5x = 0$

(ii) Prove that: $(-1 + i\sqrt{3})^4 (-1 - i\sqrt{3})^5 = 512\ w^2$

(iii) Find the quadratic equation whose roots are: $3 + \sqrt{2}, 3 - \sqrt{2}$

(iv) The area of a rectangular field is 252 square meters. The length of its side is 9 meter longer than its width. Find its sides.

(v) Solve the equation: $(\sqrt{(3x+2)} + \sqrt{x})/(\sqrt{(3x+2)} - \sqrt{x}) = 4/1$

(vi) Pressure P of gas in a container varies directly as Temperature T. When pressure is 50 N/m^2, temperature is 75°C. Find pressure when temperature is 150°.

(vii) Resolve into partial fraction: $(x^2 + 3x + 1)/((x - 1)(x^2 + 3))$

(viii) Verify De-Morgan's laws for the following sets. $U = \{x|x \in N \wedge 1 \leq x \leq 20\}$ $A = \{2,3,5,7,11,12,13,17\}$ $B = \{1,4,6,8,10,14,16,18\}$

(ix) Domain of a binary relation $R = \{(x,y)|y = 2x\}$ is the set $\{0,4,8\}$, find range of R.

(x) Calculate the Range, Variance and Standard Deviation for the following discrete data: x: 5, 10, 11, 13, 15; y: 2, 3, 4, 1, 5

(xi) Prove that: $(1 - \sin\theta)/\cos\theta = \cos\theta/(1 + \sin\theta)$

(xii) A lighthouse is 150 m above the sea level. Angle of depression of a boat from its top is 60°. Find the distance between the boat and lighthouse.

Section-C

Maximum Marks: 24

> Note: Attempt any three (3) questions. All questions carry equal marks.

Q3. Prove that: A straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord.

Q4. Prove that: Equal chords of a circle subtend equal angles at the centre.

Q5. Prove that: If a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point.

Q6. Circumscribe a square about a circle of radius 5 cm